

Sports Medicine: Overuse Injuries

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Session Objectives

- Contrast the pathophysiology of bone, muscle and tendon overuse injuries.
- Explain how to diagnose an overuse injury.
- Explain how to treat an overuse injury.
- Interpret common clinical scenarios of an overuse injury.



Overuse Injury Risk Factors

- To properly identify, diagnose, and treat an overuse injury, the clinician must first assess the associated risk factors for injury
- Extrinsic factors
- Intrinsic factors

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Overuse Injury Risk Factors

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Extrinsic factors – outside the individual

- Training errors;
 - Excessive volume or intensity, rapid increase, change in type, inadequate technique or recovery time
- Surface
 - Hard, soft, dynamic
- Equipment
 - Shoes – worn out, inappropriate
- Environment
 - hot, cold, humid



Overuse Injury Risk Factors

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Intrinsic Factors – Originate within

- **Malalignment**
 - Check patient for foot posture, patella heights, hip angle, bow leggedness
- **Anatomical irregularities**
 - Leg length discrepancy, muscle imbalance, weakness
- **Flexibility**
 - Generalized muscle tightness, focal areas of thickening, restricted ROM
- **Other**
 - Genetic, endocrine, metabolic

Categories of Overuse Injuries

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- Bone Stress
 - Stress reaction / Stress fracture
- Muscle Overuse
 - Focal tissue thickening and fibrosis
 - Chronic compartment syndrome
 - Muscle soreness “DOMS”
- Tendinopathy



Stress Reaction / Fracture

- Makes up ~10% of all athletic injury
- 95% of stress injury occurs in the LE
- Peak Incidence age 18-25
- Inverse relationship between lean body mass and stress reaction
- Women 1.2-10x more likely to suffer a stress injury

Skeletal Injury

Tibial stress fracture – Makes up ~50% of all diagnosed stress fractures

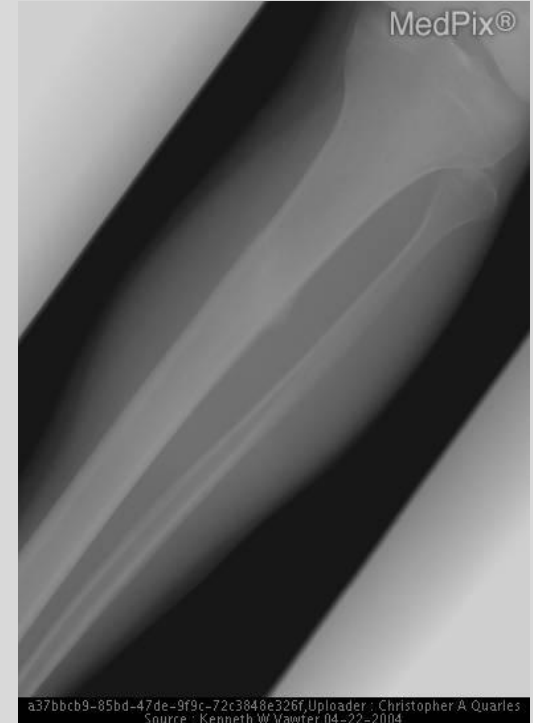
- Typically secondary to running
- Two possible locations
 1. Between the middle and distal third of the tibia along the posteromedial border – Associated with *running*
 2. Along the middle third of the anterior cortex – Associated with *Jumping* activities (basketball, dancing, gymnastics)

CURRENT Diagnosis & Treatment Family Medicine Chapter 39: Common Upper & Lower Extremity Fractures

W. Scott Black; Robert G. Hosey; Joshua R. Johnson; Kelly Lee Evans-Rankin; Wade M. Rankin

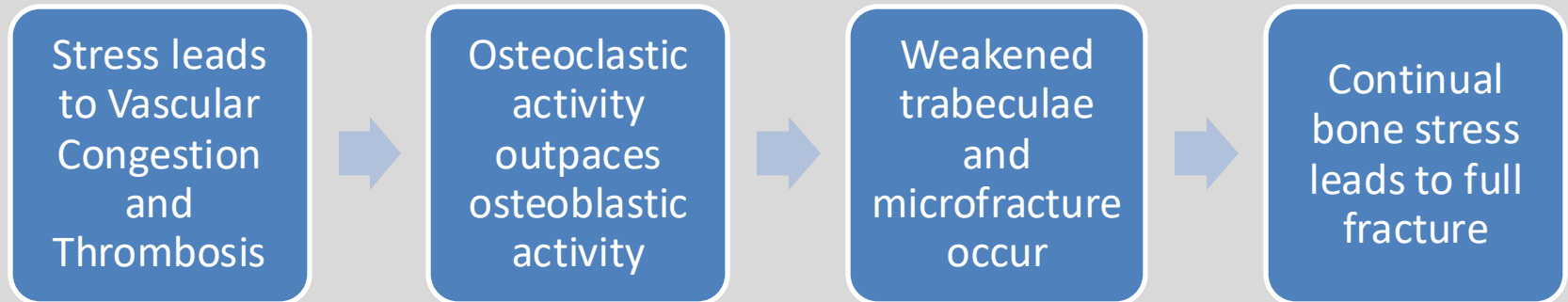
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Pathophysiology

- *Initiation* is unclear, but it is theorized that stress injury is due to muscle fatigue around a given bone leading to excessive force transmission to that bone.
- After initial bone injury, a remodeling response occurs



Skeletal Injury

Extrinsic Risk Factors of Stress Reaction / Fracture

- Changes in exercise regimen;
 - Runners who increase mileage too quickly
 - Increased intensity
 - Introduction of a new activity
 - Change in training surface
 - Change in training equipment

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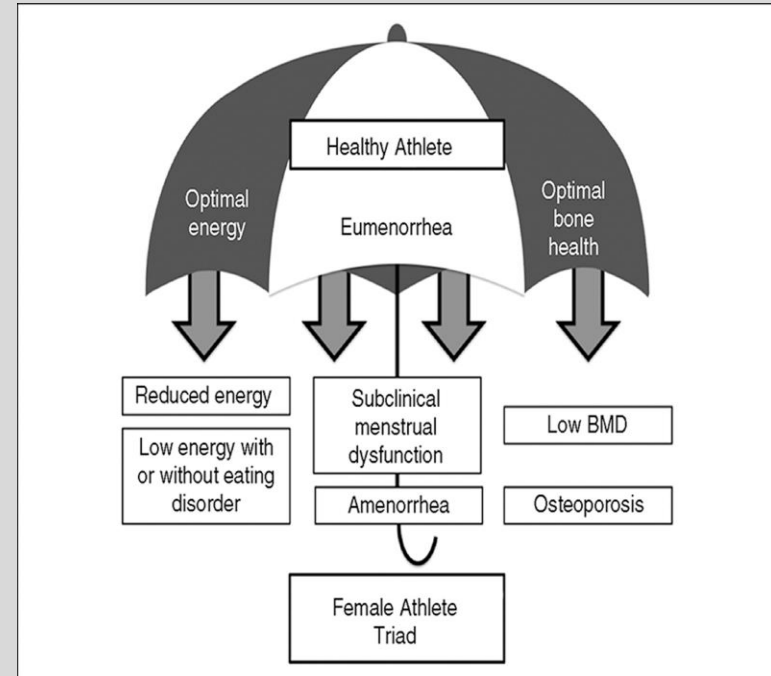
Skeletal Injury

- Intrinsic Risk Factors of Stress Reaction/Fracture
 - Low bone density
 - Low BMI
 - Dietary deficiency
 - Hormonal imbalance
 - Sleep Deprivation
- Increased risk in women
 - Female athletic triad: Amenorrhea, Disordered eating and Osteoporosis

Source: JAAOS - Journal of the American Academy of Orthopaedic Surgeons 23(7):424-432, July 2015.

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Skeletal Injury

- Clinical Presentation:
 - Typically occurs in the lower extremity due to repetitive weight bearing
 - Athletes, specifically runners are at highest risk for stress reaction / fracture
 - **Crescendo effect:** Patients report days to weeks of dull but growing pain eventually leading to inability to perform an activity
 - Initially, pain decreases with rest, but as time progresses, pain returns with less and less activity
 - Finally, pain may be present with very minimal activity and bothersome at night

Skeletal Injury Diagnosis

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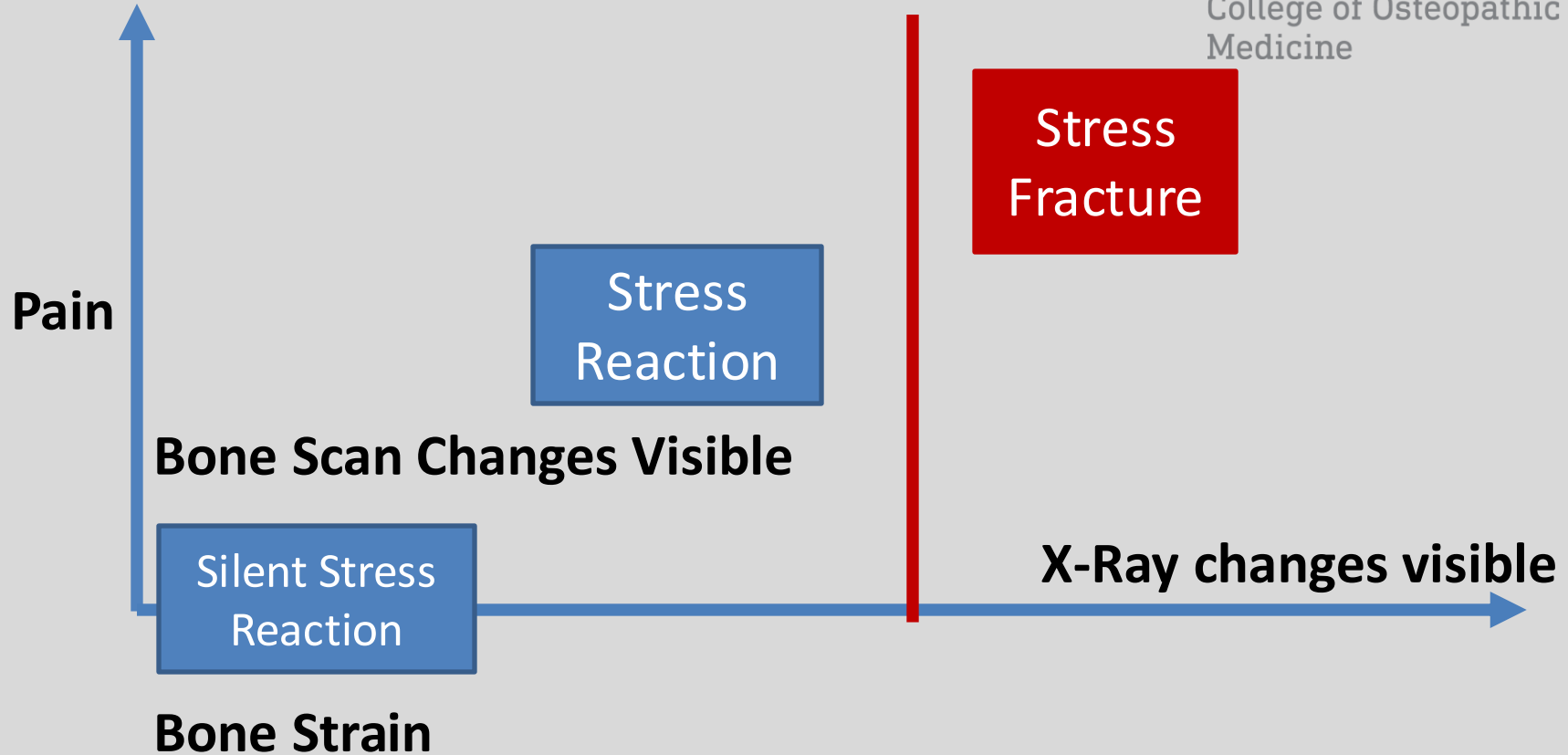
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- Clinical – based on patient presentation
- Imaging – Confirmation
 - Plain radiograph: Low sensitivity but high specificity. May only be positive after symptoms present 4-6 weeks
 - Positive signs include;
 - Periosteal reaction, Intramedullary sclerosis, Obvious fracture line
 - Technetium Bone Scan: High diagnostic power for stress fracture. Can be positive within 48 hours of symptom onset
 - **MRI- best test to confirm diagnosis**

Skeletal Injury

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Skeletal Injury

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Clinical Feature	Bone Strain	Stress Reaction	Stress Fracture
Local Pain / Tenderness	No	Yes	Yes
X-Ray Abnormality	No	No	Yes – abnormal periosteal reaction or cortical defect
MRI Abnormality	May show increased high signal (+/-)	increased high signal	increased high signal + cortical defect
Bone Scan Abnormality	Increased Uptake	Increased Uptake	Increased Uptake

Skeletal Injury

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Stress Fracture Treatment

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General Principles

- #1 if it hurts, DO NOT DO IT
- Typically, 6 weeks of *relative* rest are required
- Bracing has been shown to decrease time to return to competition by 20 days
- Avoid NSAIDs – may hinder bone healing
- Maintain fitness with cross training throughout rest period
- Correct identified precipitating factors
 - Muscle imbalance (strength training), mechanical abnormalities (orthotics), dietary planning, and training modification

Muscle overuse

- Typically secondary to muscle imbalances
- Imbalances lead to changes in muscle length and strength between agonists and antagonists – effecting overall muscle function
- This leads to overall functional and biomechanical problems leading to injury
- Muscle injuries include
 - Focal tissue thickening and fibrosis
 - Chronic compartment syndrome
 - Muscle Soreness

Muscle Overuse

Focal tissue thickening and fibrosis

- Repetitive micro trauma of overuse
 - Damage of muscle fibers
- Damage leads to the formation of **adhesions** between muscle and **cross linkages** in the fascia
- These cross linkages and adhesions may be palpated as focal areas of tissue thickening
 - Taut, thickened band arranged in the direction of stress or as large area of increased muscle tone

Focal tissue thickening and fibrosis

- Focal areas of thickening may be responsible for
 - Pain
 - Predisposition to injury of tendons due to a lack of tissue elasticity under stretch or eccentric load
 - Decreased ability for muscle to contract and relax rapidly
- Treatment for focal tissue thickening and fibrosis
 - Soft tissue therapy (OMT)
 - Strengthening
 - Stretching

Muscle Overuse

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Overuse

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Physical Therapy

Gross Muscle/Fascial Tightness

Microscopic Muscle Damage

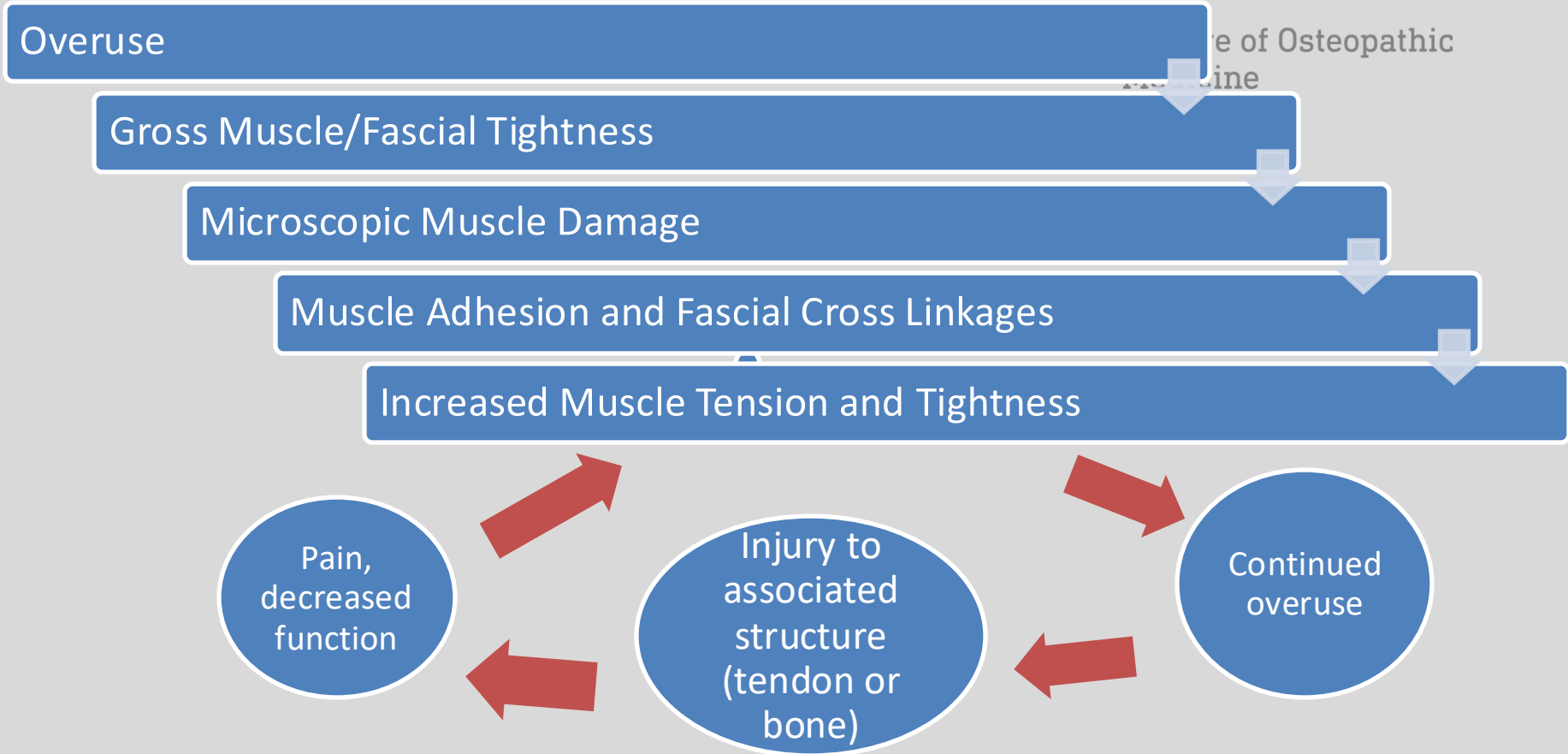
Muscle Adhesion and Fascial Cross Linkages

Increased Muscle Tension and Tightness

Pain,
decreased
function

Injury to
associated
structure
(tendon or
bone)

Continued
overuse



Chronic Compartment Syndrome

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Muscle overuse - Intermittent and reversible pathologic elevation of compartment pressure following exertion

- Most common in the lower leg
 - Can occur in forearm (tennis, rock climbing, weight lifting)
- Compartments
 - Created by relatively inelastic fascial sheaths of collagenous tissue



Chronic Compartment Syndrome

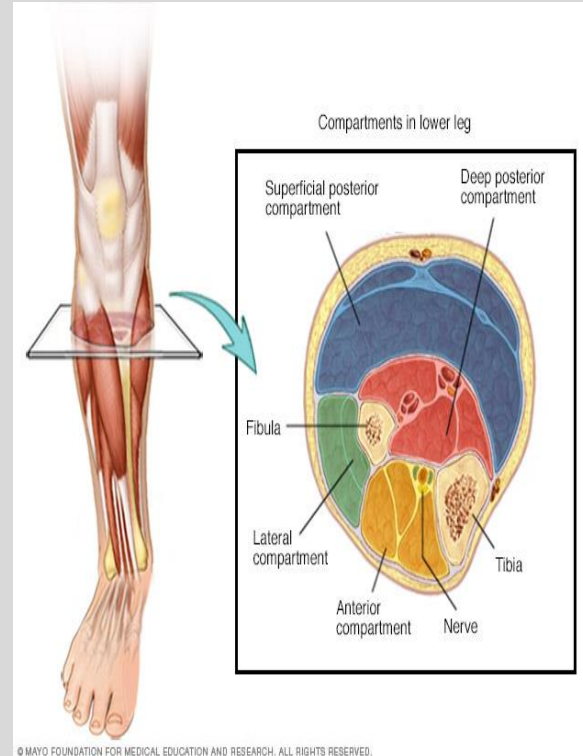
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- **Exercise** raises intracompartmental pressure, may cause local muscle swelling, accumulation of fluid in the interstitial space
 - Tight fascia prevents adequate expansion of the compartments
- Increased pressure leads to **impaired blood flow** and pain on exertion
- **Compression of neurological structures** may also contribute to pain
- Excessive muscle hypertrophy may also be involved in compartment syndrome

Compartments:

- **Anterior:** (*dorsiflexors*) tibialis anterior, extensor hallucis longus, extensor digitorum longus peroneus tertius
- **Lateral:** (*plantarflexion / eversion*) Peroneus brevis and longus
- **Deep posterior:** (*plantarflexion/inversion*) Tibialis posterior, flexor digitorum longus, flexor hallucis longus
- **Superficial Posterior:** (*plantarflexion*): gastrocnemius, soleus, plantaris

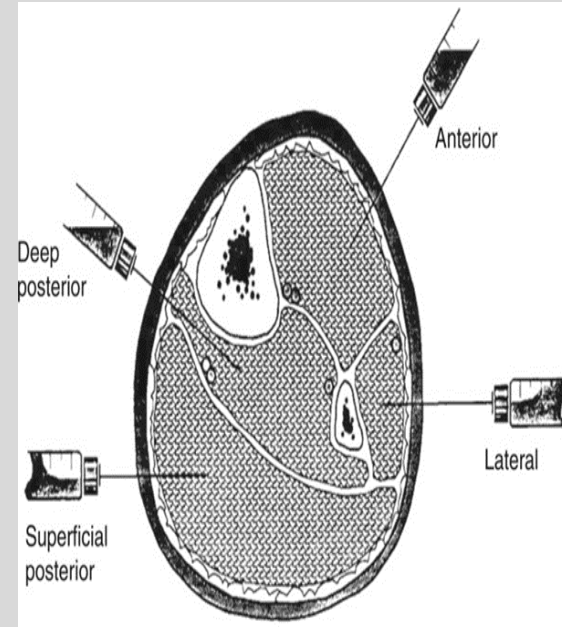


Chronic Compartment Syndrome

- **Clinically:** The main symptom is pain and pressure that commences with activity and dissipates with rest
- **Measurement of compartment pressures**
 - Can be done at rest, or during pain provoking exercise
 - Catheter placed into muscle compartment suspected to cause pain
 - First exercise muscle until symptom reproduction.
 - Next measure compartment pressure for 5 minutes.

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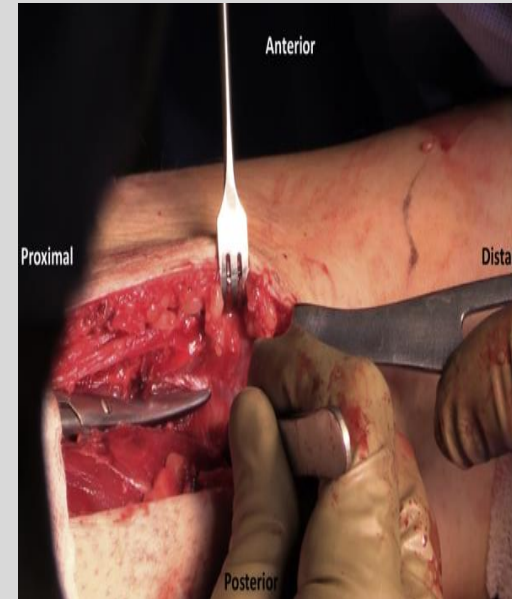


Chronic Compartment Syndrome

- **Treatment**
- Soft tissue therapy
 - OMT, stretching exercises
- Correction of biomechanical abnormalities
 - Muscle strengthening, correction of form, physical therapy
- **Fasciotomy** – release of fascia surgically

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Source: Dai, A. Z., Zacchilli, M., Jejuriar, N., Pham, H., & Jazrawi, L. (2017). Open 4-Compartment Fasciotomy for Chronic Exertional Compartment Syndrome of the Leg. Arthroscopy techniques, 6(6), e2191-e2201.. doi: [10.1016/j.jeats.2017.08.014](https://doi.org/10.1016/j.jeats.2017.08.014)

Muscle Soreness

DOMS – Delayed Onset Muscle Soreness

- Pain **24-48** hours *after* exercise activity
- Typically in untrained individuals or when attempting a new exercise
- More severe after **eccentric** (muscle contraction while lengthening) movements
- Thought to be due to muscle fiber damage leading to cell membrane swelling and resultant;
 - Decrease in muscle force production
 - Increase in passive tension and local swelling

DOMS

Treatment

- Active recovery
- Stretching
- Compression sleeves
- NSAIDs
- Massage
- OMT
- Ice bath – constricts blood flow / decreases damage

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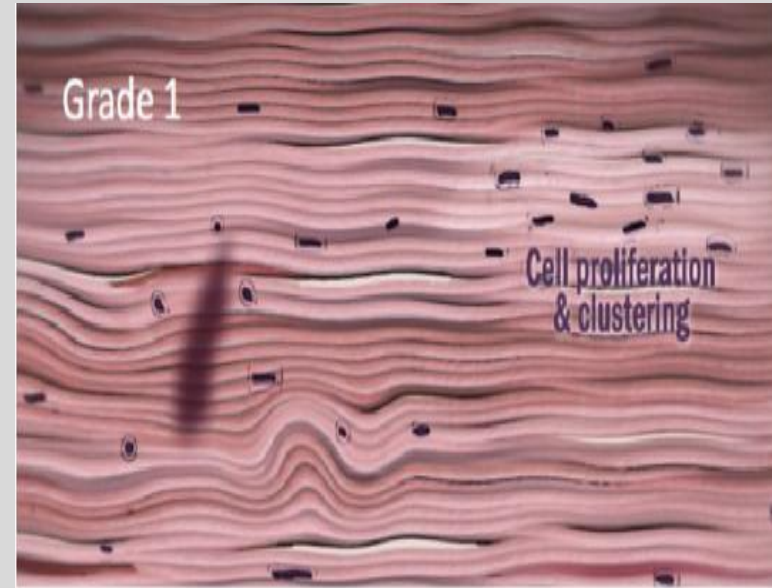


Tendinopathy

- Clinical presentation
 - Pain with use or in the morning localized to tendon
 - Typically pain free with rest
 - **Typical pain pattern:** Pain begins with exercise, but decreases as patient continues to warm up, then returns as patient cools down
 - **Physical exam:** Local tenderness and thickening of the associated tendon

Stage 1 – Reactive Tendinopathy

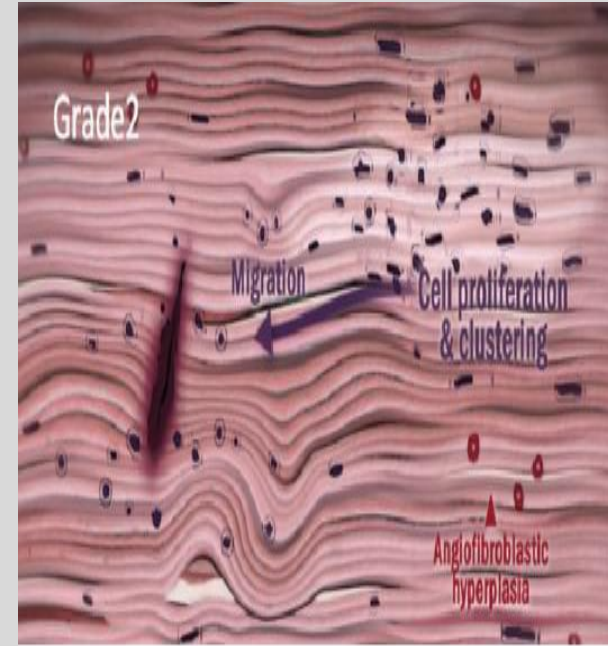
- Non-inflammatory response of tendon cells to tensile or compressive overload
- Tendon cells activate and proliferate, producing repair proteins (proteoglycans)
- Proliferation → thickening of tendon



Source: Bhabra, G.S., Wang, A.W., Ebert, J.R., Edwards, P.K., Zheng, M., & Zheng, M.H. (2016). Lateral Elbow Tendinopathy. *Orthopaedic journal of sports medicine*.

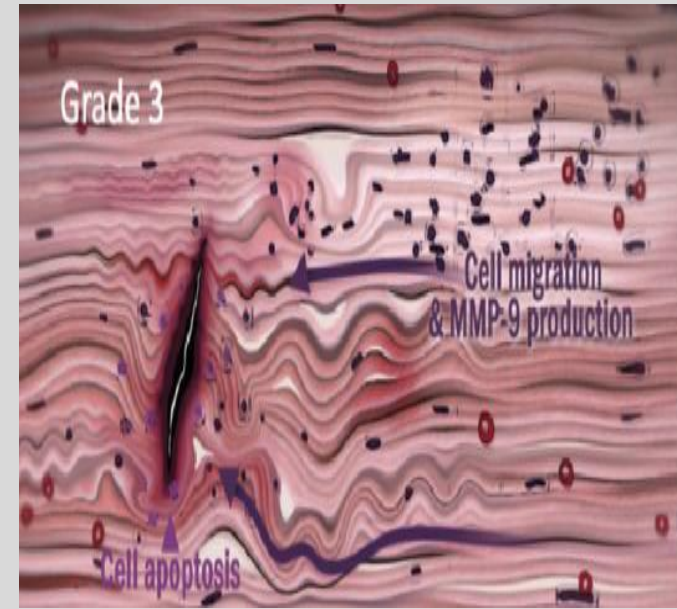
Stage 2 – Tendon Dysrepair

- Increased tendon matrix breakdown
- Tendon cells continue to predominate and become chondrocytic (rounded appearance), myofibroblasts begin to proliferate
- Further increased production of proteoglycans and collagen leading to disorganization of matrix
- Disorganized matrix allows for ingrowth of new blood vessels and nerves



Stage 3 – Degenerative tendinopathy

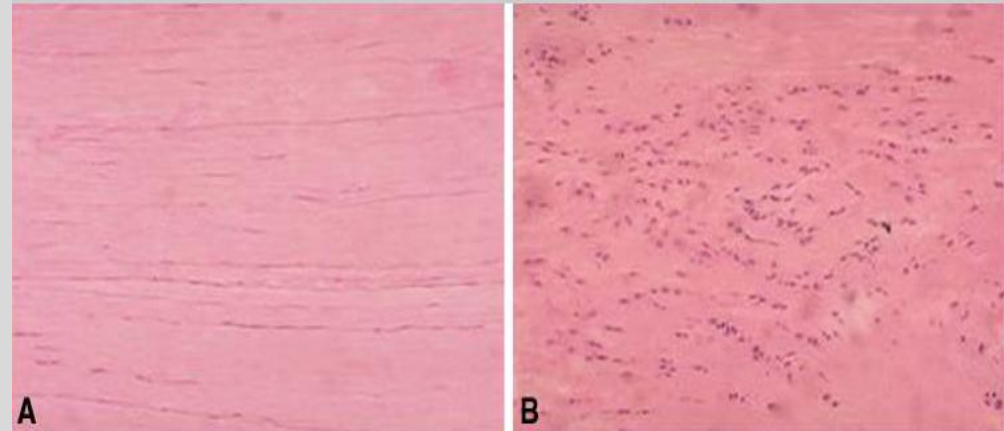
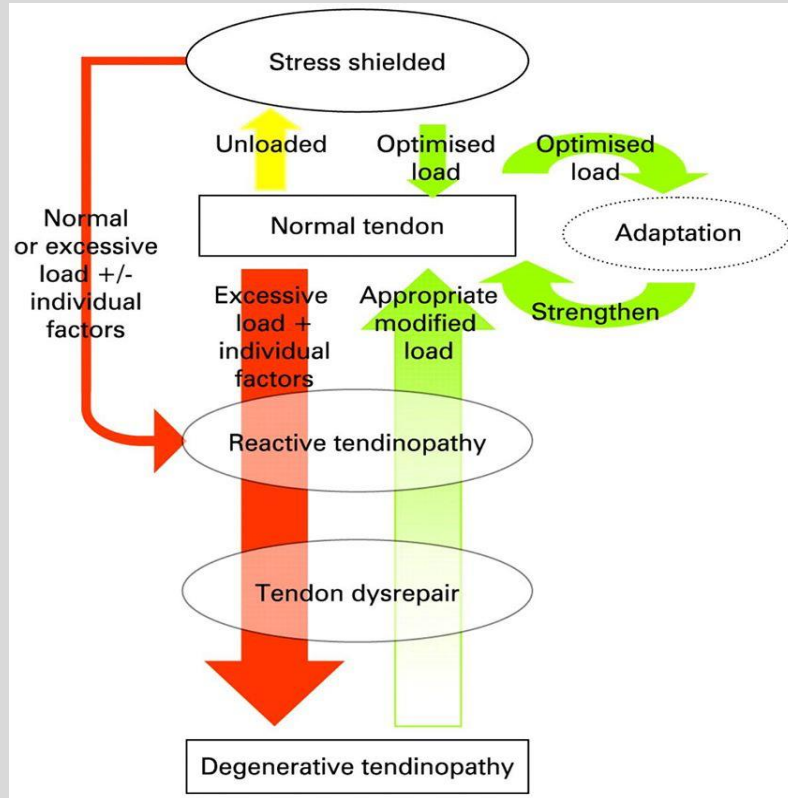
- Progression of tendon degeneration to evident apoptosis of normal cells
- Large areas of matrix disorganization replaced by neovascularization and matrix breakdown products – little remaining normal collagen



Source: Bhabra, G.S., Wang, A.W., Ebert, J.R., Edwards, P.K., Zheng, M., & Zheng, M.H. (2016). Lateral Elbow Tendinopathy. *Orthopaedic journal of sports medicine*.

Tendon Overuse

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A - Normal tendon has parallel, longitudinal architecture with scattered elongated tenocytes

B - Tendinopathy tendon shows disorganized collagen architecture, high cellularity of rounded tenocytes

Tendon Overuse

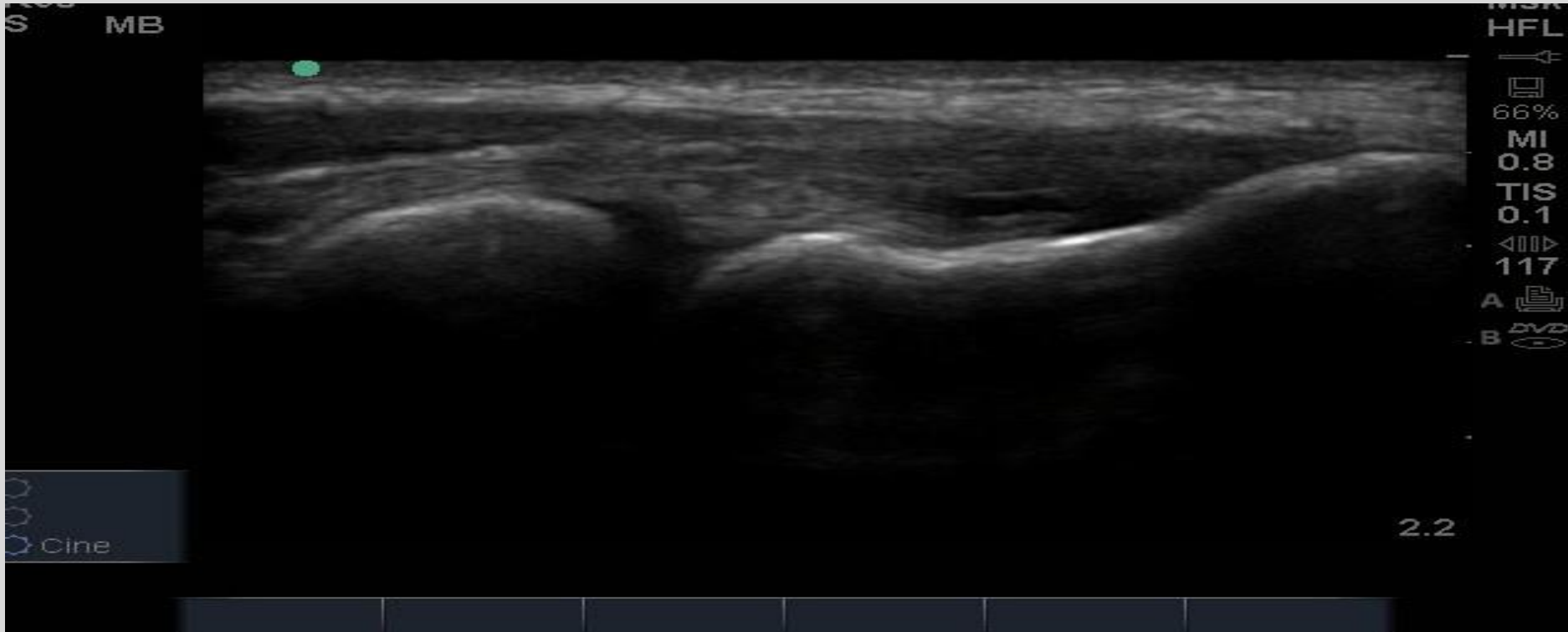
- **Diagnosis and Treatment**
- **Diagnosis:**
 - Clinical – pattern of pain
 - Imaging - Ultrasound hypoechogenicity, increased sized and thickness of tendon
- **Treatment:**
 - Conservative: Physical therapy, stretching, activity modification, change of footwear, injections (avoid steroids due to tendon rupture risk)
 - Operative: Tendon augmentation or tendon transfer (FHL transfer to Achilles)

Case - Mr. AG

- A 53 yo male
- Non-radiating achy right elbow pain at lateral aspect x 2 months.
- Gradual
- 7/10 today
- Worsens with bowling which he does 3-4 times per week.
- Denies numbness, tingling, swelling or redness.

Case Mr. AG

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Source: <https://theultrasoundsite.co.uk/common-extensor-tendon-tear/>

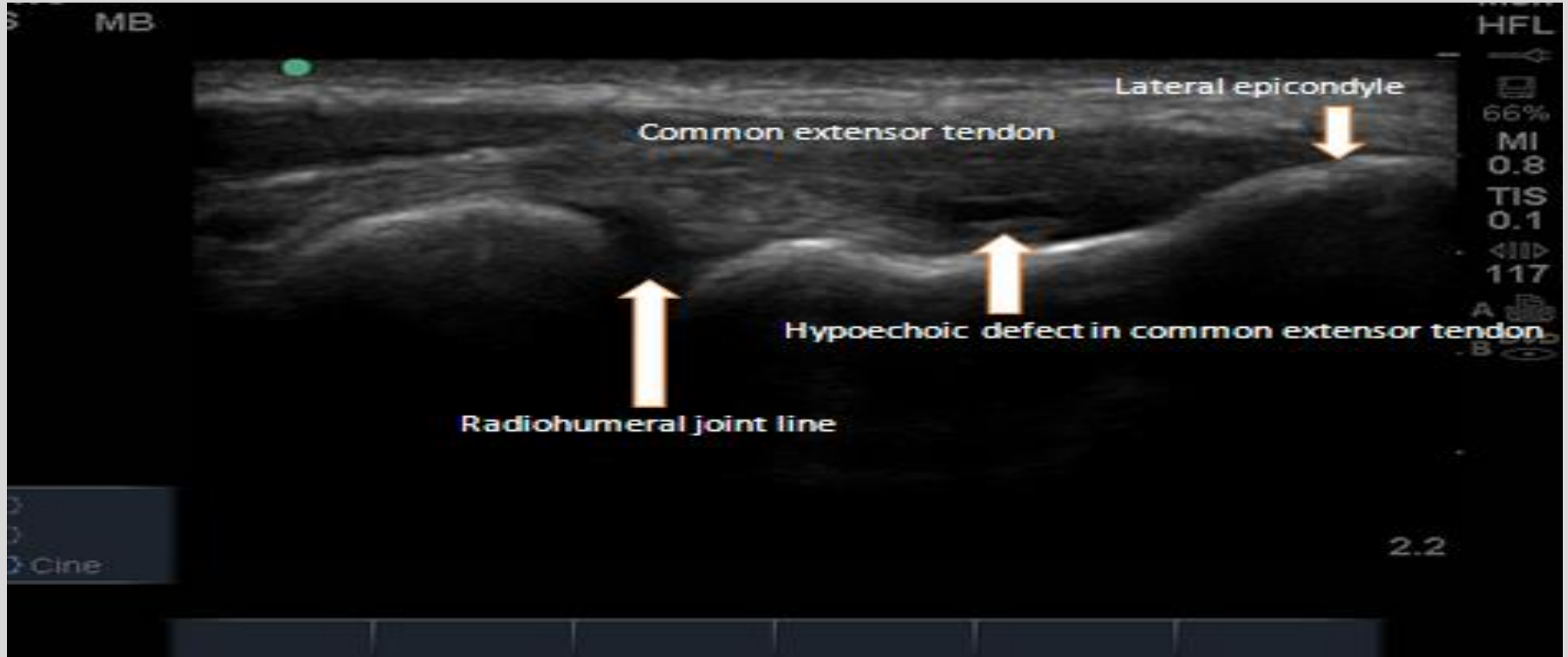
Case – Mr. AG

Which of the following is the most likely diagnosis?

- A. Biceps tendinosis
- B. Medial epicondylitis
- C. Radial tunnel syndrome
- D. Lateral epicondylitis
- E. Radiocapitellar chondromalacia

Lateral Epicondylitis

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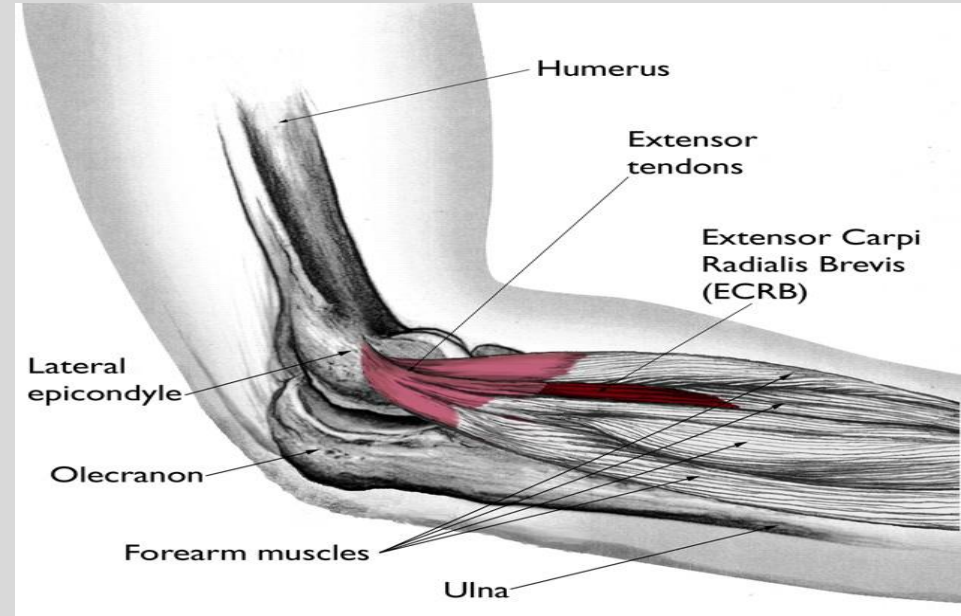
Source: <https://theultrasoundsite.co.uk/common-extensor-tendon-tear/>

Lateral Epicondylitis

- Chronic tendinosis
- Extensors
- Racquet sports
- Pain
- Prevention

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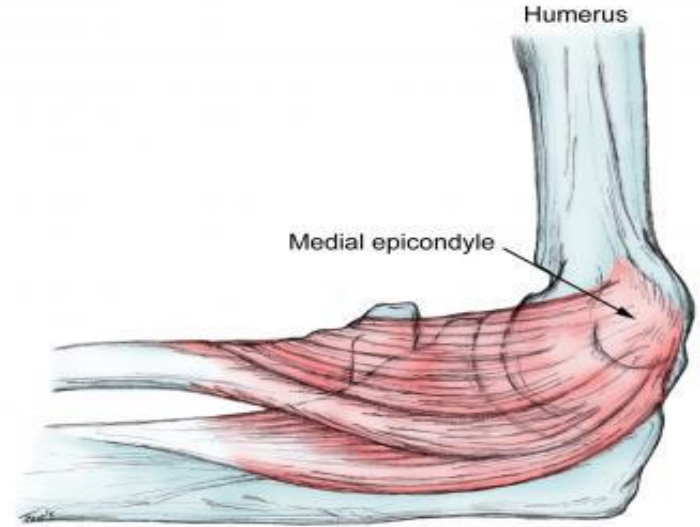


Medial Epicondylitis

- Pronator teres and flexor carpi radialis.
- Flexion and pronation
- Pain
- Golf/throwing
- Ulnar nerve

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Medial (inside) view of the right elbow

A 44-year-old patient presents to the clinic complaining of intermittent right forearm pain and pressure of three months duration. Pain is achy and tight with occasional numbness that is present when they plays tennis but subsides immediately after. What is the most likely diagnosis?

- A. Stress fracture
- B. Tennis elbow
- C. DOMS
- D. Chronic compartment syndrome
- E. Muscle strain

A 44-year-old patient presents to the clinic complaining of intermittent right forearm pain and pressure of three months duration. Pain is achy and tight with occasional numbness that is present when they plays tennis but subsides immediately after. Which of the following is the best initial treatment?

- A. Fasciotomy
- B. Fasciectomy
- C. Compartment pressure measurement
- D. Casting
- E. Physical Therapy

A 22-year-old male presents to the clinic complaining of right shoulder pain radiating to the anterior upper arm. On physical exam when the right shoulder is flexed and externally rotated with thumb pointed up and the patient is resisting downward pressure the pain is reproduced. What would you expect to find on histopathology?

- A. Lymphocytes
- B. Neovascularization
- C. Increased osteoclast activity
- D. Organized tendon
- E. Muscle necrosis

A 22-year-old male presents to the clinic complaining of right shoulder pain radiating to the anterior upper arm. On physical exam when the right shoulder is flexed and externally rotated with thumbs pointed up and the patient is resisting downward pressure the pain is reproduced. Which of the following treatments should be avoided?

- A. Relative rest
- B. Physical therapy
- C. OMM
- D. Corticosteroid injection

Summary Slide

- Contrast the pathophysiology of bone, muscle and tendon overuse injuries.
- Explain how to diagnose an overuse injury.
- Explain how to treat an overuse injury.
- Interpret a common clinical presentation of an overuse injury.
- Core References:
 - Brukner, P., & Khan, K. (2016). *Clinical sports medicine*. McGraw Hill. Chapters 5 & 13.

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